The Post-API Age

Week 3 · Foundations module · Textbook Chapter 3

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This week at a glance

The web used to invite us in. That has changed in recent year.

Why did this bargain change? *How* can we keep doing public interest research anyway?

Monday | Concepts

- The golden age
- Enclosure, exemption & erosion
- Counter-values, lineages & misconceptions

Wednesday | Notebook lab

- Measure data access and forensic analysis of dead endpoints

Friday | Textbook revisions

- Reviewing issues and proposing pull requests

Monday lecture

Read Chapter 03 of the Web Data Science book

Wednesday lab

Complete **Recommended Exercises:** fill in the empty cells, run everything, export to HTML, upload to Canvas by **Sunday 11:59pm**

Friday notebook

You do not need to wait until Friday to begin to submit issues from textbook or notebook, upload link to issue by **Sunday 11:59pm**

1. Monday · Concepts

The golden age of web data access

Between roughly 2008 to 2018 the web was a mostly open book that was easy(ish) to study

- Websites rendered a mix of static and dynamic content
- Social platforms shared data: Twitter in 2006; Facebook in 2010; Reddit's API was free; Google exposed trends
- Platforms traded data access for lock-in to their ecosystems, credibility, recruitment
- This built my field of computational social science and a generation of data journalism and civic tech

When researchers today say the **"post-API age"** (Freelon 2018), this is the era they measure against.

They just gave out data for free!

Three pressures

(1) Enclosure, (2) Exemption, (3) Erosion

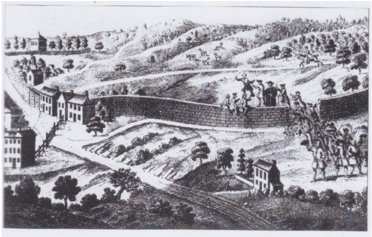
Pressure 1 — Enclosure

Enclosure privatizes a shared resource — traces back to the fencing in of English common land. The data still exist, just on the other side of a paywall.

- **Twitter/X:** a free academic archive (2006–2021) → discontinued; enterprise access starts at $42,000/month
- **Reddit:** free since 2008–2023 → $0.24 per 1,000 calls; 7,000+ subreddit protests and Pushshift cut off

Two drivers:

1. **Independent scrutiny** — researchers and journalists answering uncomfortable questions about polarization, harassment, privacy, financials, *etc.*
2. **AI training data** — uncontaminated user-generated content becomes enormously valuable for training AI models



This land is my land now, it isn't your land anymore

Textbook Ch. 3, "Enclosure." Facebook's CrowdTangle shutdown (2024) was the template.

Pressure 2 — Exemption

Exemption: platforms operate beyond outside scrutiny while running the same studies internally at will.

Every A/B test is a behavioral study, and internal teams run them at will — while outside researchers meet:

- cease-and-desist letters citing the Computer Fraud and Abuse Act;
- Terms of Service that prohibit the very collection research needs (Fiesler et al. 2020);
- account bans — e.g., Meta disabling NYU's Ad Observatory (2021), justified as *protecting privacy*.

When "privacy" restricts only accountability research — never ad targeting on the same users — it is functioning as exemption, not protection.

Consenting users' data, still shut down.

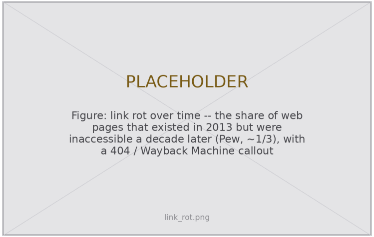
Textbook Ch. 3, "Exemption." Courts push back at the margins (*Sandvig*, *Van Buren*, *hiQ*; see Ch. 2). EU DSA Article 40 may mandate vetted-researcher access.

Pressure 3 — Erosion

Erosion is the quietest pressure: enclosure and exemption are things platforms *do*; erosion is what happens to the web *itself*.

- **Links rot.** About half the links in U.S. Supreme Court opinions are dead; a 2024 Pew study found about a third of 2013 web pages gone a decade later.
- **Platforms churn.** GeoCities vanished; Parler went offline overnight; Twitter became X and broke a decade of embedded links.
- **Sites redesign** and silently break every scraper — fragility you will feel in Ch. 6 & 8.

For research this is a **reproducibility crisis:** a study on the Twitter Academic API cannot be rerun at any price (Davidson et al. 2023).



The record decays unless someone saves it.

Textbook Ch. 3, "Erosion." Two later chapters are counter-erosion: archives (Ch. 7) and automation (Ch. 14).

Activity: which pressure hits your source?

Pick a platform or dataset you would actually want for a project — ideally one you are considering for the final project.

1. Name what you would collect, concretely: which records, which fields, how far back.
2. Decide which pressure is most likely to close it on you, and why.
3. Find one piece of evidence *right now*: a pricing page, an API changelog, a developer-forum complaint, a dead third-party tool.

Report back in pairs; we will tally the pressures on the board.

The three suspects

- **Enclosure** — paywalled, repriced, licensed away
- **Exemption** — bans, cease-and-desists, ToS traps
- **Erosion** — link rot, shutdowns, redesigns

Keep your pick

Step 4 of this week's take-home asks you to add two endpoints *you* care about to the access matrix. Use this one.

Counter-value 1 — Openness

Three values guide public-interest work inside the post-API age. They are a stance for working under the pressures, and each has a practice attached.

Openness: conduct your own work so that it resists enclosure.

- Document *how* you collected, beyond what: endpoints, parameters, dates, and the platform's constraints at collection time (Gebru et al. 2021). That record lets a future researcher understand your dataset after the source changes or vanishes.
- Practice redundancy: deposit data in durable repositories (Zenodo, institutional archives), lean on the Internet Archive and Common Crawl, prefer openly licensed sources.
- Wikipedia is the standing counter-example to enclosure — open API, open license, full public dumps (Ch. 10).

Openness in one habit

Record the date. Your access matrix is a snapshot; the date is what turns it into evidence someone else can build on.

Textbook Ch. 3, "Openness."

Counter-value 2 — Oversight

Oversight: consequential systems should be inspectable by parties who do *not* own them.

- Independent audits of platform algorithms and content moderation; documentation standards that make datasets and methods legible to review.
- Regulation converts access from a favor into an obligation: EU DSA Article 40 compels vetted-researcher access to very large platforms.
- Oversight is the direct answer to exemption — the A/B-test asymmetry, closed from the outside.

This course is oversight

Someone who can retrieve, parse, and analyze web data independently is someone a platform cannot simply refuse. The skills taught here are oversight skills.

Textbook Ch. 3, “Oversight.”

Counter-value 3 — Ownership

Ownership: prefer arrangements where communities govern the infrastructure their data lives in.

- Federated systems on open protocols — ActivityPub (Mastodon), AT Protocol (Bluesky) — distribute control: no single acquisition or pricing decision can enclose the whole network (Ch. 12).
- Community archives, data cooperatives, and volunteer efforts like ArchiveTeam hold collective traces collectively.
- The long-horizon value: hardest to practice, and the durable answer to enclosure.

The trade

Federation exchanges convenience for control: smaller populations, uneven moderation, more work per query. You will feel both sides of that trade in Ch. 12.

Textbook Ch. 3, "Ownership."

Lineage 1 — Journalism

Other professions met this problem long before us — powerful actors controlling information the public needs — and built institutions in response. Five lineages, five lessons.

- Freedom-of-information laws exist because reporters and reformers spent decades establishing that **government records belong to the public**.
- The muckraking tradition established that **private power, too, is a legitimate target** of sustained scrutiny — Standard Oil, the meatpacking industry.

The echo in data science

You work in this lineage every time you file a records request for data an agency will not publish (Ch. 11) — one of the "four more doors" coming up.

Textbook Ch. 3, "Public Interest Lineages."

Lineage 2 — Law

- **Discovery** compels parties to produce documents they would prefer to hide, on the theory that adjudication without evidence is theater.
- **Due process** demands that decisions affecting people be explainable and reviewable — a proceduralist answer to arbitrary power.

The echo in data science

Algorithmic accountability inherits the due-process demand almost verbatim: explainability and review, aimed at software. Week 2's cases (*Sandvig*, *hiQ*) were this lineage in court.

Textbook Ch. 3, "Public Interest Lineages."

Lineage 3 — Accounting

- After the 1929 crash, mandatory audits and standardized reporting turned corporate finances **from private mystery into public record**.
- The machinery that makes disclosure trustworthy at scale: **standards, professional norms, and independent examiners** — goodwill does not audit itself.

The echo in data science

Datasheets, model cards, and dataset documentation standards are early drafts of the same machinery for data and models (Gebru et al. 2021).

Textbook Ch. 3, “Public Interest Lineages.”

Lineage 4 — Urban planning

- Environmental impact statements and public comment periods encode a principle: decisions about **shared resources** require **advance disclosure and community input**.
- The affected get to speak *before* the bulldozer moves, and the record of what was said is public.

The echo in data science

Algorithmic impact assessments copy the template: disclose before deploying, and take comment from the people the system will govern.

Textbook Ch. 3, "Public Interest Lineages."

Lineage 5 — Engineering

- Licensure, safety codes, and ethics canons — many written in the aftermath of collapsed bridges and exploded boilers — established that practitioners owe **duties to the public that override duties to employers.**
- An engineer can refuse to sign off, and the refusal has institutional backing: a license the employer cannot revoke.

The unfinished echo

Data science is still writing its equivalent: no license to lose, no canon with teeth — yet. You are entering the field while its canon is being drafted.

Textbook Ch. 3, “Public Interest Lineages.” Missing Manual Ch. 8: “Artifacts Have Politics.”

What this means for practice — the course roadmap

The rest of this course equips you with a diverse toolkit precisely because no single method of access is reliable.

When this closes...	...we go here
APIs shut down or reprice	scrape the HTML — Ch. 6 (static), Ch. 8 (dynamic)
pages change or vanish	web archives — Ch. 7
records locked inside PDFs	document extraction — Ch. 9
platforms restrict automation	federated alternatives — Ch. 12
one-off pulls stop sufficing	scheduled automation — Ch. 14

The strategic lesson is **redundancy and flexibility:** skills across the full range, so the closure of any single source cannot shut down your research program.

Textbook Ch. 3, “What This Means for Practice.”

Four more doors — beyond the toolkit

1. **Official research access programs** — the Meta Content Library, TikTok's Research API, Reddit's researcher program. Real but rationed: narrow eligibility, publication restrictions, revocable access. Use them, and document their constraints in your methods.
2. **Data donation** — users hold export rights to their own data and can volunteer it. Donation studies and instrumented browser extensions convert individual data rights into collective observation, with consent built in.
3. **FOIA and public records requests** — disclosure law reaches data no API offers: inspection records, communications, contracts. Ch. 11 covers what agencies publish proactively.
4. **Crowdsourcing and citizen science** — when no one holds the data, communities create it: distributed observation and collaborative annotation, from air quality to political advertising.

Textbook Ch. 3, "What This Means for Practice."

Myth check — vote before the next slide

Five claims students arrive with. For each, commit: **true or false?**

1. "The post-API age means APIs are gone."
2. "If the API closes, I'll just scrape."
3. "If data is publicly visible, collecting it is permitted and ethical."
4. "This only matters for social media research."
5. "Nothing can be done about it."

Rules

Write your five answers down first; then we take a hand count, myth by myth. Defend your minority votes — some of these are *almost* true.

Myth check — what Chapter 3 says

1. **APIs are everywhere.** More exist than ever; what is disappearing is *free access for outside observers*, replaced by paid tiers, gated programs, and licensing deals.
2. **Scraping is one tool in a portfolio.** Ch. 6 & 8 teach it — and it carries legal ambiguity (Ch. 2), breaks when sites redesign, and scales poorly.
3. **Visibility, permission, and ethics are three different questions** (Fiesler et al. 2020). A public post's author may never have expected a dataset; a viewable page's ToS may forbid automated collection.
4. **Enclosure follows AI-training value** wherever it appears: news archives, code repositories, forums, image libraries. Government data (Ch. 11) stays open by legal mandate.
5. **The environment is contested.** Regulation is creating access obligations, archives are institutionalizing preservation, federated platforms are growing — and practitioners' choices are part of the contest.

Textbook Ch. 3, "Common Misconceptions."

Wednesday: measure it, then take it home

Monday's claims are checkable, and Wednesday we check them:

- Build an **access matrix**: probe eight live endpoints with no credentials and tabulate who answers, who refuses, and how.
- Run **dead-endpoint forensics**: autopsies of Pushshift, CrowdTangle, and Twitter v1.1 — three retirements, three different failure signatures.

Bring a laptop with the course environment working; we start from `ch-03-post-api.ipynb`.

Take-home: the 7-step audit

1. Named `User-Agent`
2. Probe one endpoint by hand
3. All eight → dated DataFrame
4. Add two endpoints *you* care about
5. Diagnose one refusal, attribute it
6. Autopsy one retired endpoint
7. Write up what changed

Deliverable

Completed notebook, exported to **HTML**, on Canvas by **Sunday 11:59pm**.

2. Wednesday · Notebook Lab

Claims you can check — the access matrix

Enclosure, exemption, and erosion are claims about the world. Claims about the world can be checked. The chapter's audit asks eight endpoints one question: *if I show up with no credentials, what happens?*

```python
def probe(name, url):
    """Ask one endpoint for data without credentials; describe the answer."""
    try:
        response = requests.get(url, headers=HEADERS, timeout=20)
    except requests.RequestException as error: # no answer is a result too
        return {"api": name, "status": None, "verdict": "unreachable"}
    verdict = ("open" if response.status_code == 200
               else "gated" if response.status_code in (401, 403) else "other")
    return {"api": name, "status": response.status_code, "verdict": verdict,
            "content_type": response.headers.get("content-type", "")}
```

Measured 28 Aug 2026: Wikipedia, Hacker News, arXiv, Mastodon, Bluesky **open**; Reddit and X **gated**; Meta Graph refuses without a token. Your run *will* differ — that is the finding, so record the date.

Textbook Ch. 3, "An Access Matrix." Mastodon is the only host that states its rate limit before you hit it.

Read the rows, not the totals

The useful information is in the differences:

- **Wikipedia, Hacker News, arXiv** answer anonymous requests with data, and have for a decade — a non-profit, an incubator, a university consortium. No one here monetizes attention.
- **Reddit**: status 403, content-type `text/html` — not an API refusing you, a *block page* served where JSON lived from 2008 to 2023. Enclosure is a content-type header.
- **X**: `401 application/problem+json` — a machine-readable, honest refusal. Differently built gates tell you whether applying for access would even help.

Attribute before you conclude

While the chapter was drafted, GitHub returned 403 — a striking finding, since GitHub allows 60 anonymous requests/hour. The *body* named the drafting sandbox, not GitHub. Check `server` and `via`, read the body, retest from a second network.

Textbook Ch. 3, "Reading the Failure, Not Just the Status Code." A measurement that describes your instrument is worse than none.

Dead-endpoint forensics — three ways to die

How a retired endpoint refuses tells you whether anything can be recovered:

- **Pushshift** answers `403` in JSON: `{"detail": "Not authenticated"}`. **Gated, not gone** — it still runs, for Reddit moderators. A gated service can be negotiated with.
- **CrowdTangle** does not resolve at all. No HTTP conversation happens. **Enclosure completed**; nothing left to ask.
- **Twitter v1.1** parses your request and rejects it (error 215). Alive, serving paying customers. **Repriced, not removed.**

The archive keeps the receipts

Wayback's last captures: `api.pushshift.io` — 25 May 2023, six weeks before Reddit's pricing took effect. `crowdtangle.com` — 14 Aug 2024, the month Meta retired it. The shutdowns are written in the archival record.

Archived docs are how you reconstruct what a five-year-old paper's dataset contained. More in Ch. 7.

Textbook Ch. 3, "Dead-Endpoint Forensics." Always pass `timeout=` — a dead host may drop packets rather than refuse.

The take-home — a guided audit in seven steps

The notebook's **Recommended Exercises** walk the audit end to end. Fill in each empty cell:

1. Define `HEADERS` with your named User-Agent
2. Probe **one** endpoint by hand; look at the answer
3. Build the eight-endpoint matrix as a DataFrame
4. Add **two** endpoints you might actually study
5. Diagnose one refusal: `server`, `via`, body
6. Autopsy one retired endpoint: gone, gated, or repriced?
7. Interpret: what differed from the book, and why?

Submitting

Run all cells, then export: *File → Save and Export Notebook As → HTML*. Upload the `.html` file to Canvas by **Sunday 11:59pm**. Canvas accepts only `.html` — code and output together, as executed.

Start in class, finish at home. Work in pairs; submit your own run.

Textbook Ch. 3, Recommended Exercises. Date-stamp your matrix — it is a snapshot, not a fact.

If you finish early — the additional exercises

The chapter's **Additional Exercises** are optional, open-ended extensions — worth your time, not required:

1. **Extend the autopsy** — two more retired endpoints: gone, gated, or repriced, and who is left to ask?
2. **Timeline** — five API-access events, three platforms, 2008 to today; tag each with its pressure.
3. **Framework application** — 500 words on a platform change *not* in the chapter.
4. **Contingency plan** — a project on one source, plus three fallbacks if it vanishes mid-study.
5. **Public interest audit** — how would current restrictions hit a real API-enabled project?
6. **5617**: read Davidson et al. (2023) and Perriam et al. (2020); write a 2–3 page research-access brief for one platform.

Fallbacks worth knowing

Official research programs (real but rationed) · data donation · FOIA / records requests (Ch. 11) · crowdsourcing.

Show-and-Tell

Bring one: an access change that surprised you, a dataset you fear losing, or a question for a research design.

Openness in practice — document the collection

The audit teaches you to date-stamp a measurement; openness generalizes that habit. Record **what** you collected and also **how**: endpoints, parameters, dates, and the platform's constraints (Gebru et al. 2021). A future reader (or you, in a year) can then reconstruct exactly what was allowed:

```
collection = {
    "source": "reddit.com (PRAW, r/ClimateChange)",
    "endpoint": "GET /r/{sub}/top params: t=year, limit=1000",
    "collected": "2026-02-14",
    "access_tier": "researcher program (application-gated)",
    "constraints": "no redistribution of raw comments; 100 req/min",
    "known_gaps": "pre-2023 history unavailable -- Pushshift offline",
}
import json, pathlib
pathlib.Path("datasheet.json").write_text(json.dumps(collection, indent=2))
# The datasheet travels next to the data; the source may change, the record does not.
```

"If the API closes, I'll just scrape."

Visibility, permission, and ethics are three different questions.

Scraping is one tool in a portfolio —
not an escape hatch that makes enclosure irrelevant.

3. Friday · Textbook Revisions

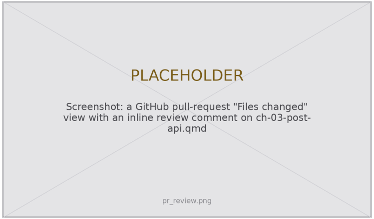
Improving Chapter 3 together

Last week you filed an **issue** — a report that something is wrong. Today you take the next step and propose the **fix** itself: your **first pull request**, against `ch-03-post-api.qmd`, plus reviews of two classmates'.

The workflow, in GitHub Desktop or `gh` (no raw `git` needed):

1. **Clone** the book repo, then create a **branch**
2. Edit the chapter; **commit** with a clear message
3. **Push**, then open a PR describing *what* and *why*
4. Review two peers' PRs: kind, specific, actionable

This is a **fast-moving** chapter — prices, lawsuits, and regulations change monthly — so it always needs hands.



Contributions & reviews count toward the Textbook Revisions grade (15%).

A revision menu for Chapter 3

High-value targets — pick one and scope it to a single PR:

- **Add the missing figures.** The chapter still has no figures. Contribute a three-pressures diagram, an API-access timeline (2008–2026), or a link-rot chart — the deck's placeholders show the need.
- **Re-run the audit.** The access matrix and forensics outputs are dated 28 Aug 2026 and *will* drift. Re-run the chapter's code, and where a row changed, update the printed output and the prose that reads it. This is the chapter's built-in revision engine.
- **Update the moving numbers.** Verify and re-cite the Twitter/X pricing, the Reddit–Google deal, and DSA Article 40's implementation status "as of mid-2026." Flag what has gone stale.
- **Resolve the editorial TODOs.** The source carries author-voice and "add citation when published" notes (the openness/oversight/ownership framing). Track down a source or draft the text.
- **Extend a Misconception or Further Reading.** Add a post-2023 source on AI-training licensing, tied to "enclosure logic is spreading."
- **Verify the cross-references.** Ch. 3 leans on many `@sec-` links (ethics, wikipedia, archives, automation, social, government). Confirm each resolves and is described accurately.

Targets drawn from the chapter's callouts, thin exercises, "Common Misconceptions," and "Further Reading."

A good revision, and a good review

A strong PR

- One focused change, clearly titled
- Explains the problem it fixes
- Builds (`quarto render`) without errors
- Matches the book's voice and notation
- Discloses any AI assistance

A strong review

- Runs / reads the change, not just skims
- Names specifics (line, claim, source)
- Separates "must fix" from "nice to have"
- Is generous — assume good faith
- Ends with a clear verdict

Next week: **Data formats** — XML & JSON, the shapes web data actually arrives in.

Key takeaways

1. The open-API era (~2008–2018) was a **strategic bargain**, not a natural state — and computational social science was built on the trade.
2. **Enclosure** converts collective data into private assets: shutdowns, repricing, and AI-training licensing deals.
3. **Exemption** is the asymmetry of scrutiny — platforms study users freely while blocking those who study platforms.
4. **Erosion** is infrastructural decay, so reproducibility is a problem you must solve *at collection time*.
5. **Access is measurable, not just arguable** — an unauthenticated probe places a platform on the open-to-closed gradient today, and *how* a dead endpoint refuses tells you whether anyone is left to negotiate with.
6. The response is **redundancy:** openness, oversight, ownership — and mastery of many access techniques so no single closure ends your research.

Reading: Textbook Ch. 3. See Freelon (2018); Bruns (2019); Lazer et al. (2020).